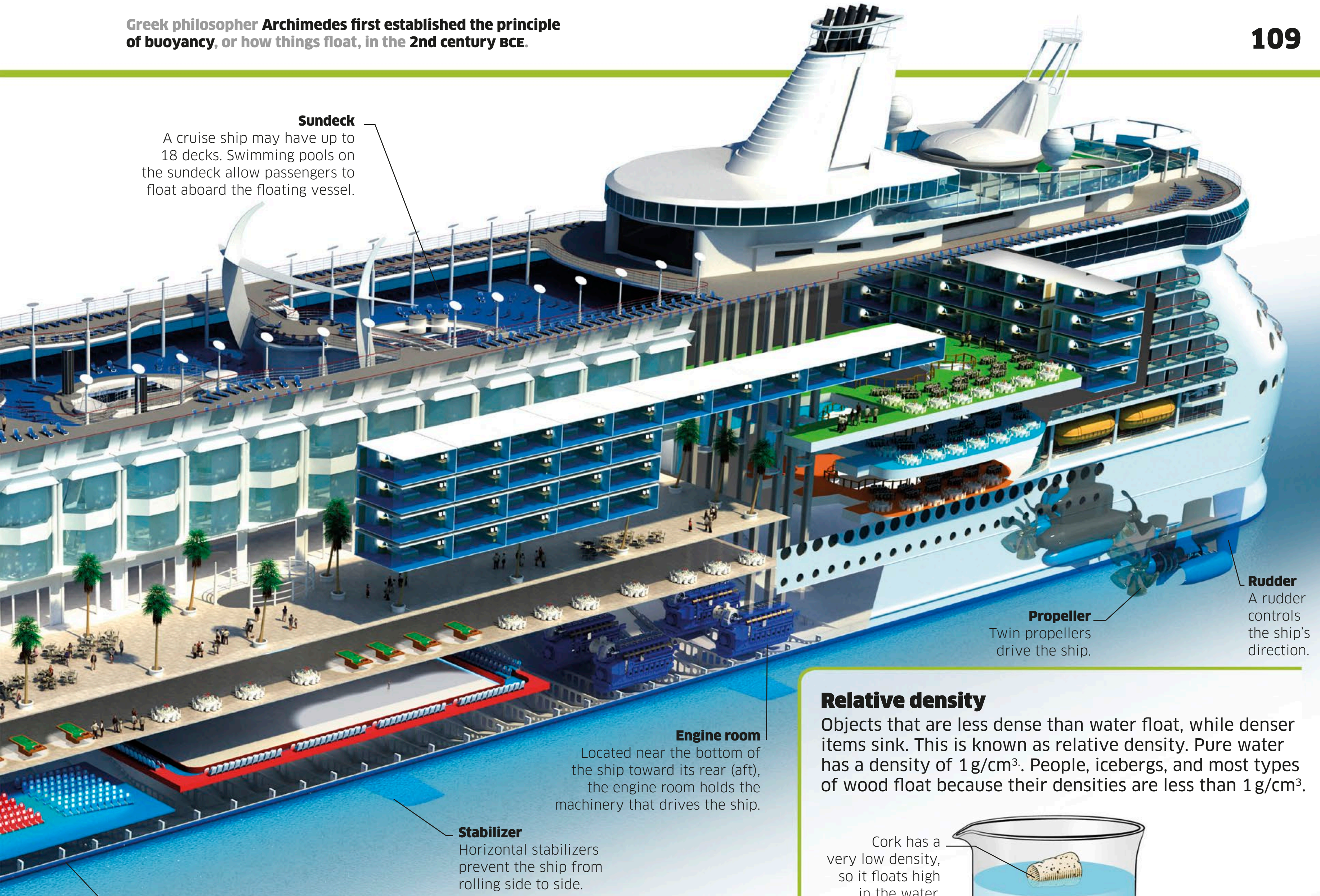
Greek philosopher **Archimedes first established the principle of buoyancy**, or how things float, in the **2nd century BCE**.

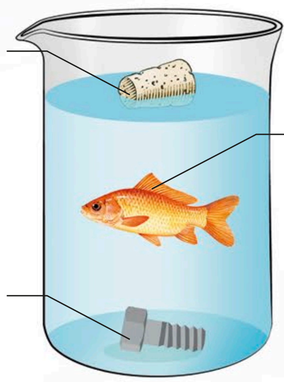


Floating city

Vast cruise ships can carry nearly 10,000 people, along with fuel, food, water, and cargo (known as dead weight), and the ship's machinery (lightweight), displacing 110,230 tons of water. How can these juggernauts of the sea float?

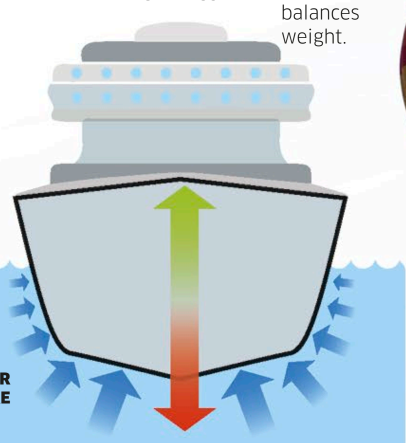
Relative density

Objects that are less dense than water float, while denser items sink. This is known as relative density. Pure water has a density of 1 g/cm^3. People, icebergs, and most types of wood float because their densities are less than 1 g/cm^3.



How boats float

Water exerts pressure on any object immersed in it. Pressure increases with depth, so the pressure on the underneath of an object is greater than the pressure on the top. The difference results in a force known as upthrust, or buoyancy. If the upthrust on a submerged object is equal to the object's weight, the object will float.



Sink or swim

A solid block of steel sinks because its weight is greater than upthrust, but a steel ship of the same weight floats because its hull is filled with air so its density overall is less than the density of water.

Floating in air

Like water, air exerts pressure on objects with a force called upthrust that equals the weight of air pushed aside by the object. Few objects float in air because it is light, but the air in hot-air balloons is less dense than cool air.